# MC Fundamentals Task Sheet

## 2D Excavator Basic Operation

### Setup and Operation


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# Learning Objective

## Goal of this Guide

In this guide you will learn 2D setup and operation within Earthworks via practical machine time application on an excavator.

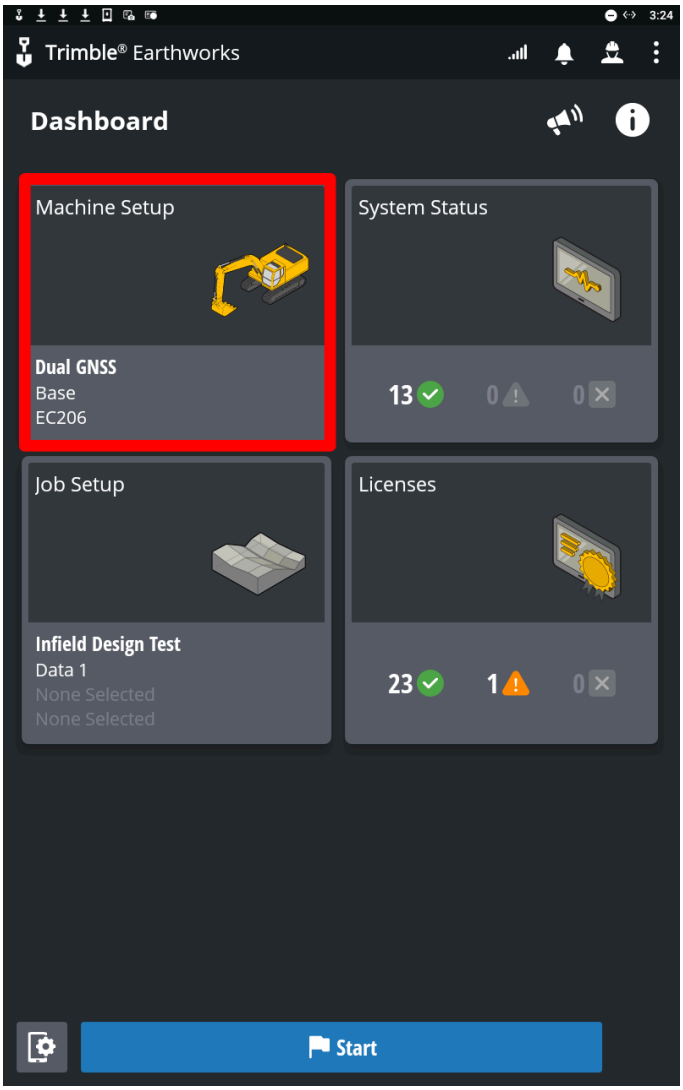
# Theory of Operation

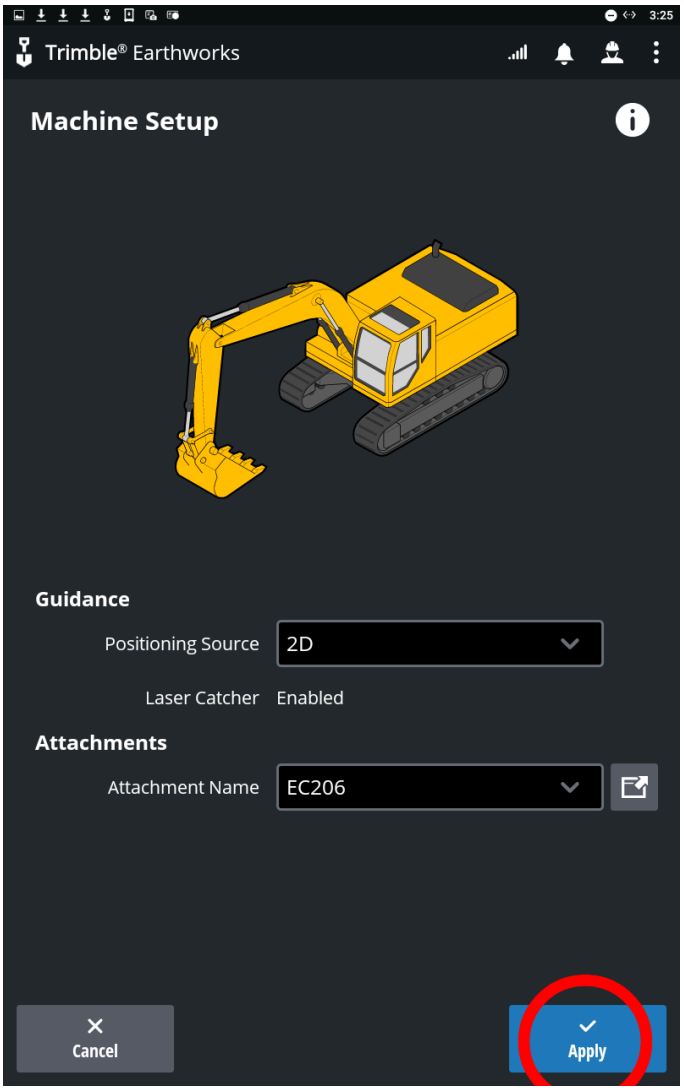
- To generate guidance information, the system measures the cutting edge's position relative to a physical reference surface or laser plane. This is called Depth and Slope guidance.
- Depth and Slope mode lets you use slopes, sections, and offsets as required, to define your target grade.
- Unlike Design guidance, depth and slope guidance does not rely on knowing the machine's three dimensional location.

# Workflow Overview

# Machine Setup

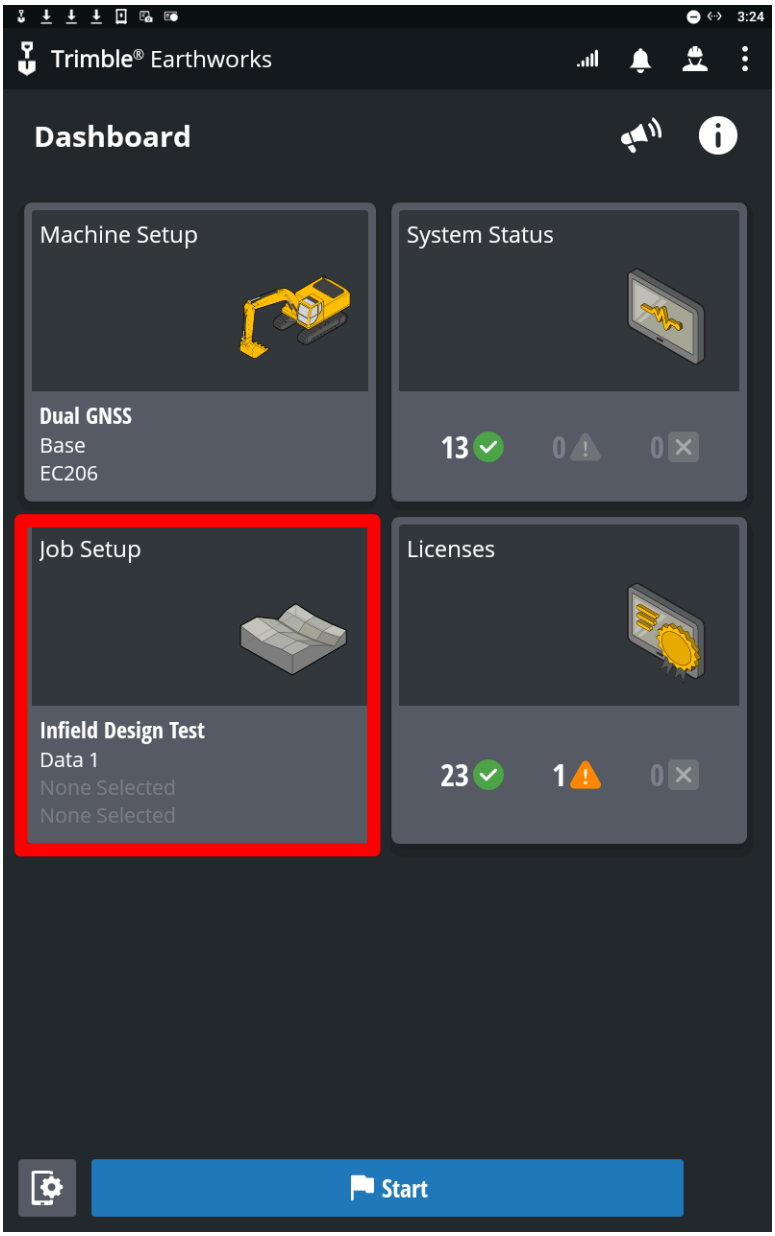
1. From the Dashboard tap the Machine Setup Tile.
2. In the Positioning Source drop down menu select **2D**.
3. Tap Apply



# Job Setup

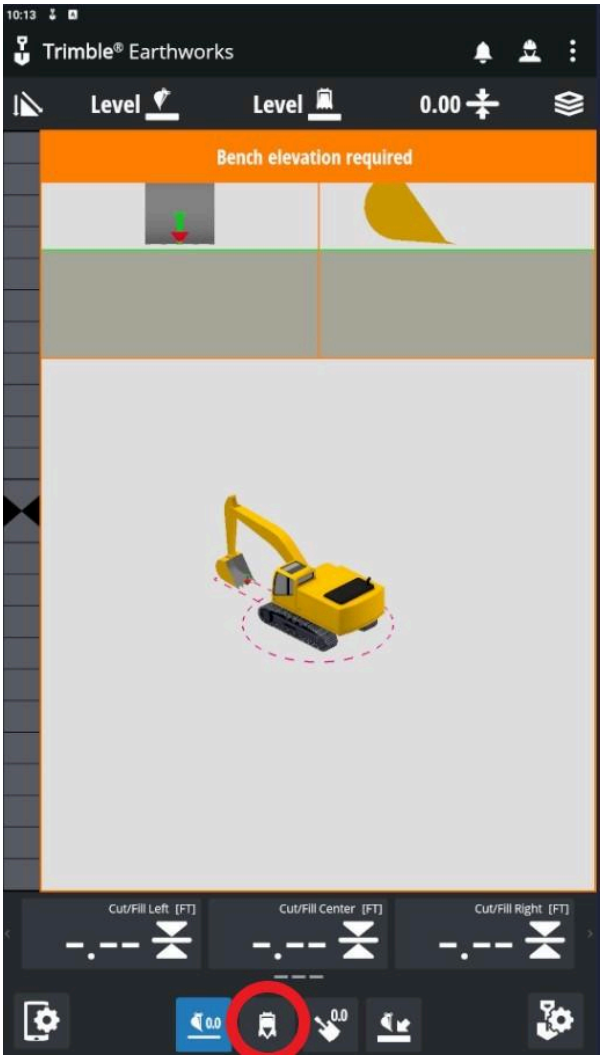
1. From the Dashboard, tap the Job Setup tile.
2. To create a new Project tap the + button next to the Project tile
3. You can also select an existing Project from the drop down menu.
4. Tap Apply.
5. From the Dashboard, tap Start. The work screen appears.

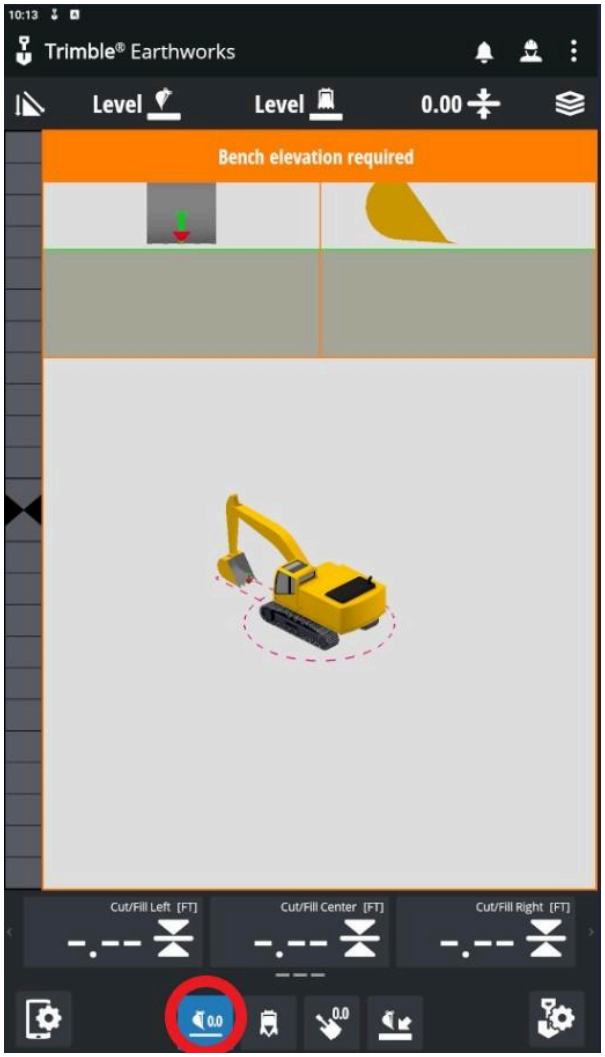


# Setting a Bench Height

Instead of going into the Bench Elevation menu, benching can be completed through the work screen.

1. Select Focus Point
2. Place your Focus Point on the elevation you would like to Bench to.
3. Tap Bench



# 2D Depth and Slope

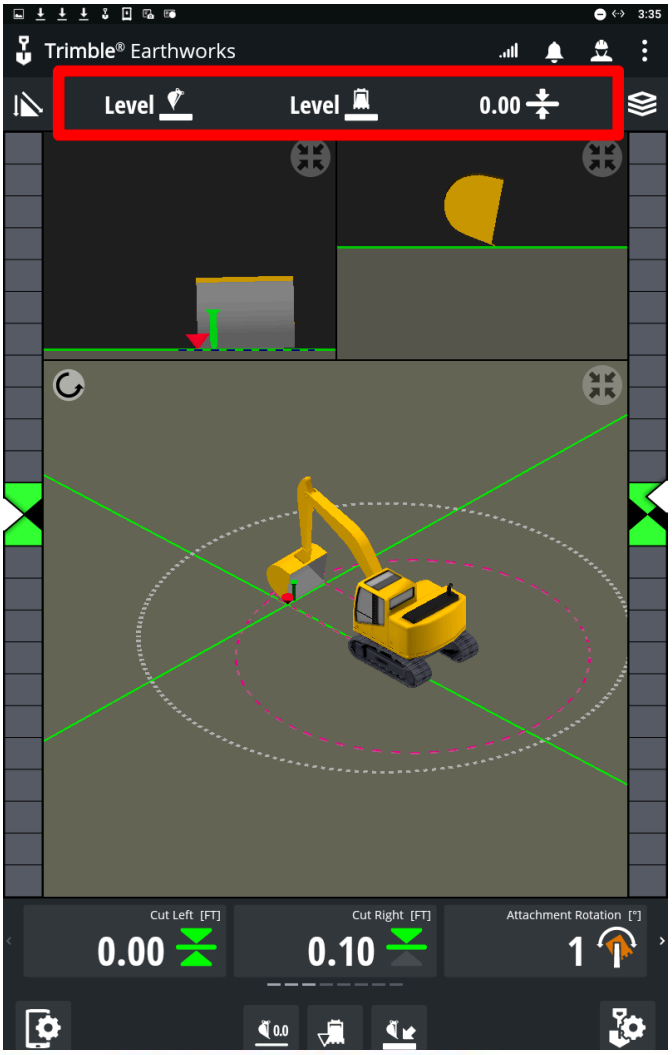
Depth and Slope mode lets you use slopes, sections, and offsets as required, to define your target grade.

To generate guidance information, the system measures the cutting edge's position relative to a physical reference surface or laser plane. This is called 2D Depth and Slope guidance. Depth and Slope mode lets you use slopes, sections, and offsets as required, to define your target grade. Unlike Design guidance, depth and slope guidance does not rely on knowing the machine's three dimensional location.

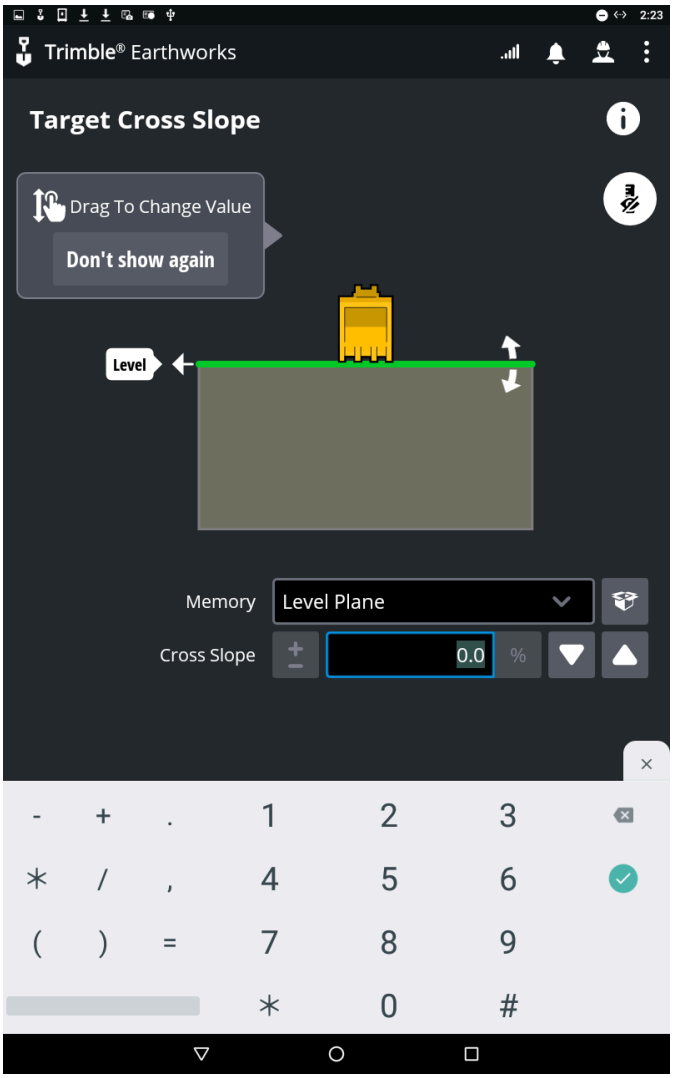
While in 2D Depth and Slope you are able to create Level Planes, Sloping Planes, and Sloping Planes with Cross Slope. The selected Mainfall and Cross Slope angles can be displayed in the upper text ribbon.

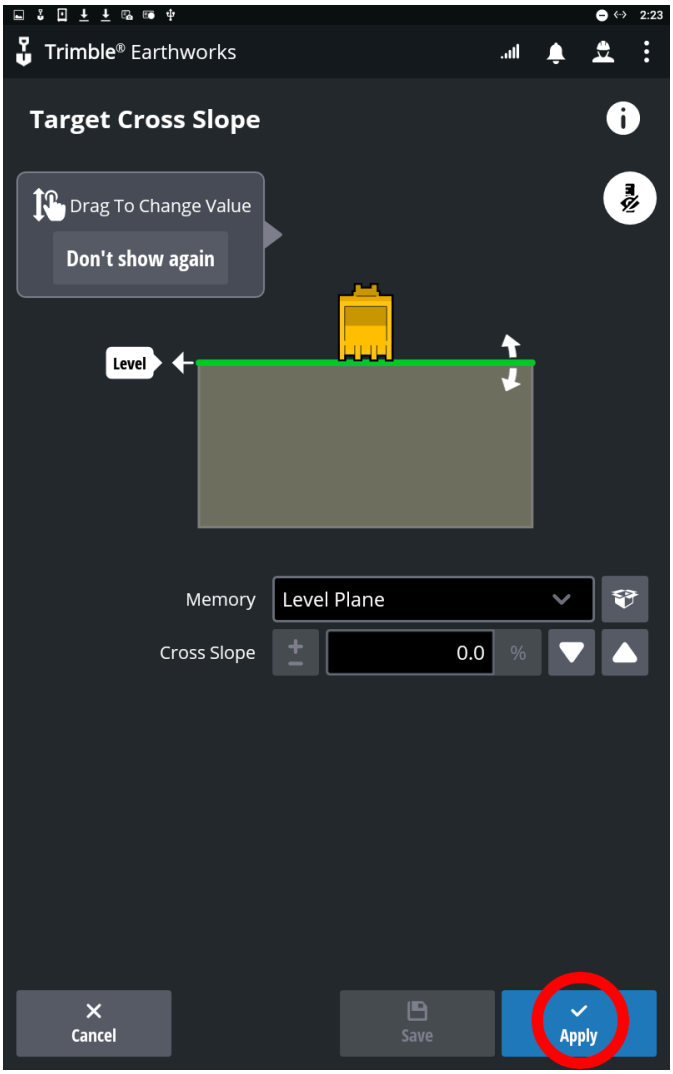
Creating Slopes in 2D

1. Press and hold the desired Tile (Mainfall or Cross Slope).



2. There are a few ways to adjust the angle for the Slopes
    a. Drag the green line up or down to increase or decrease the value.
    b. Use the Up/Down arrows to increase or decrease the value.
    c. Manually enter the value with the on screen keypad by touching the numeral box.
    d. You can flip the offset from positive to negative with the +/- button.
3. Tap Apply

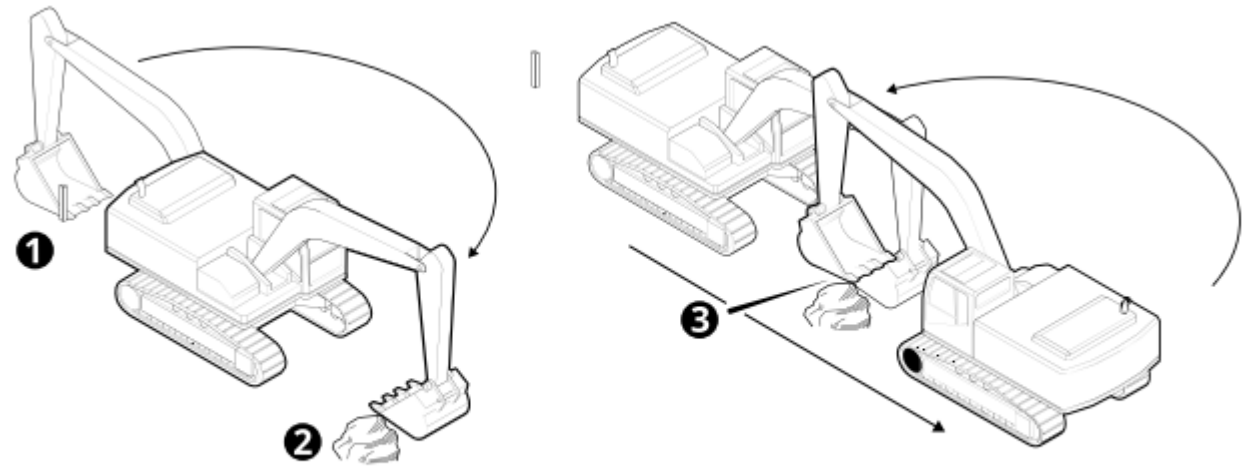


# Transferring a Bench

You can use Touch Point to transfer a bench elevation via a common hard point when you use Depth and Slope mode without a laser catcher. Touch point is only available after you bench the system and requires a common feature, such as a rock, that you can easily access from both your original working position and your new working position.

*Note: If you do not have a heading sensor you cannot rotate the cab when you move from one working position to another and must reach the Touch Point in the digging orientation of the excavator front linkage.*




1 Bench elevation. 2 Set the Touch Point 3 Regain the Touch

1.Position the attachment focus on the Touch Point (common point).

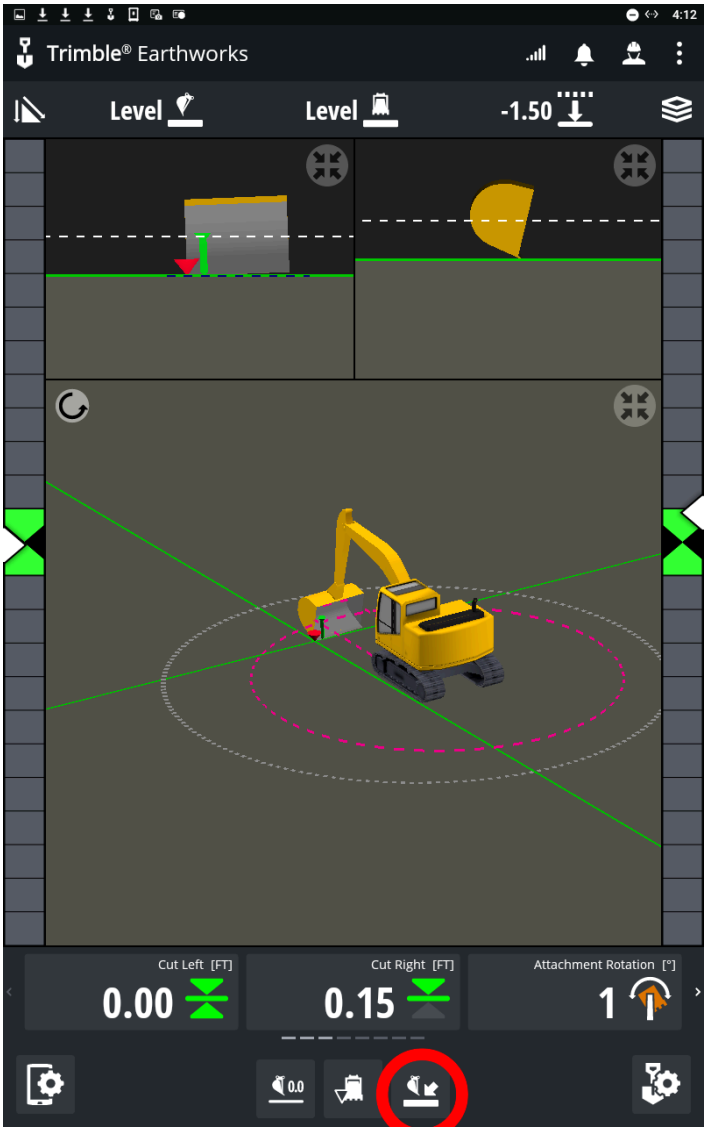
2. Tap the Touch Point icon in the Text Ribbon.

3. Move the machine to your new working position and place the attachment focus on the same Touch Point (common point).

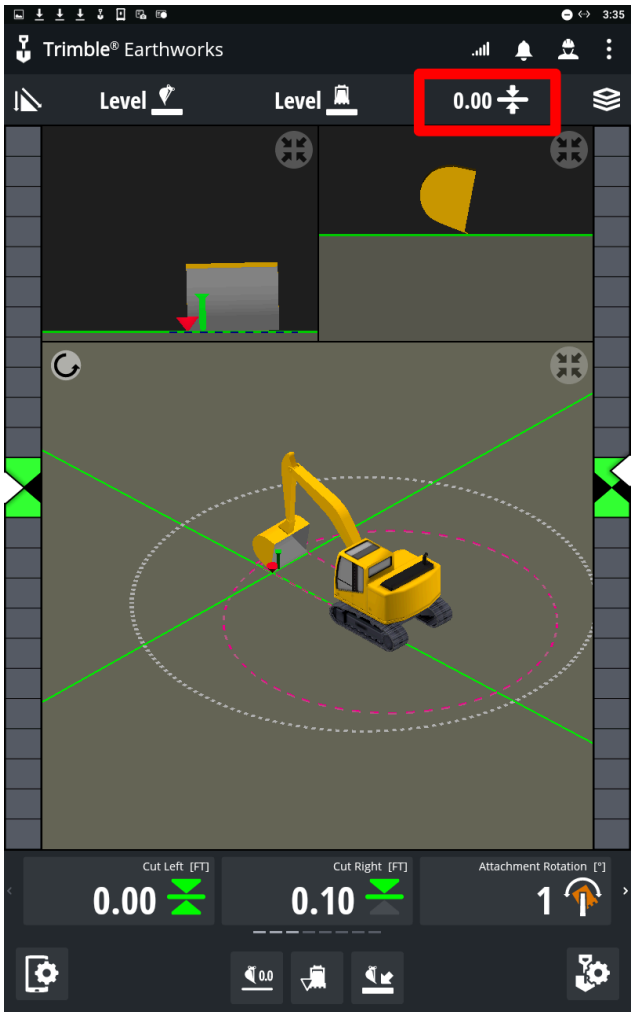
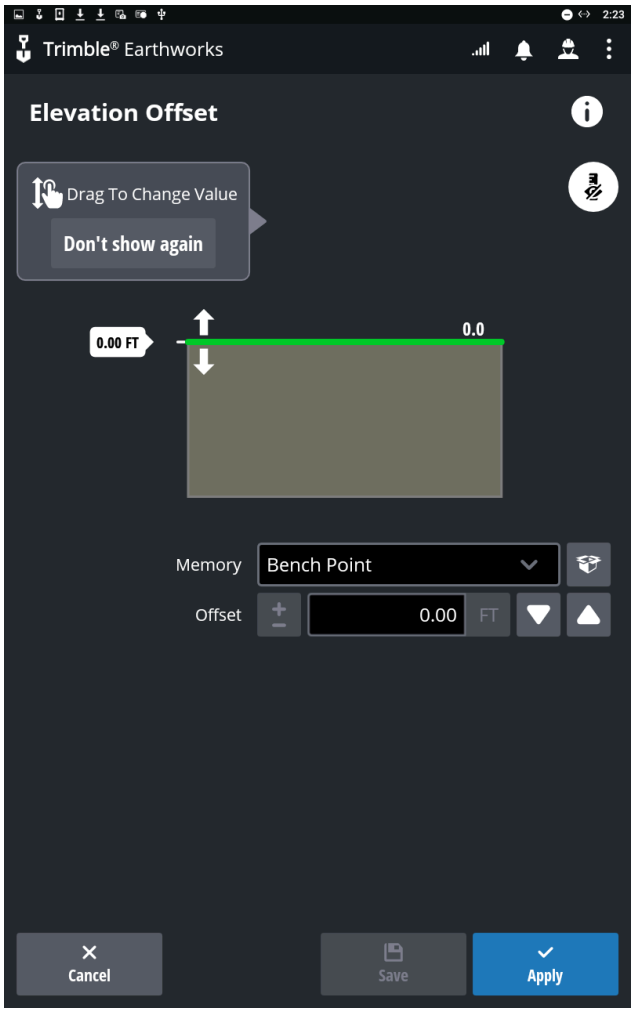
7. Tap Regain. The reference elevation is transferred.

8. Tap Done. You can now work the surface relative to the original bench.

***Note-*** *If you do not have a heading sensor you cannot rotate the cab when you move from one working position to another and must reach the Touch Point in the digging orientation of the excavator front linkage.*



# Elevation Offset

After a benchmark height has been established, offsets can be created to help reach grade. Multiple offsets can be stored in the memory and selected at a later time. Tapping on the Elevation Offset Tile will cycle through your created offsets, you can also press and hold and select one from the Memory drop down menu.

1. Press and hold the Elevation Offset Tile
2. There are a few ways to adjust the Elevation Offset.
    a. Drag the green line up or down to increase or decrease the value.
    b. Use the Up/Down arrows to increase or decrease the value.
    c. Manually enter the value with the on screen keypad by touching the numeral box.
    d. You can flip the offset from positive to negative with the +/- button.
3. Tap Apply